FORWARD DEPLOYED AI ENGINEERING

Capstone Project Report

An intelligent customer support system for CloudServe Solutions

Prepared by: Kaushik Dutta

Evaluation run: 2 November 2026 — one run, 80 visible validation tickets

Version: 1.0

This report documents the implementation and its measured limits. It is not a deployment approval. Personal effort records, reflection, independent answer ratings, provider-live verification and the hidden assessment set remain outstanding.

Submission filename: KaushikDutta_Capstone_Report.pdf

Contents

Section	Purpose
1. Executive Summary	Decision and measured result
2. Problem	Client context and scope
3. Discovery Findings	Interview and dataset synthesis
4. Requirements	Traceable functional and quality requirements
5. Architecture and Design	Pipeline and key decisions
6. Implementation	Components, API and operations
7. Evaluation	Method, metrics, interpretation and limitations
8. Governance and Risk	Controls, fairness and incident response
9. Requirements Revision	Evidence-driven changes
10. Conclusions	Readiness and next actions
Appendices A–D	Metric detail, traceability, risk/AI-use and submission checklist

The supplied submission guide asks for a 20–30 page report excluding appendices, with these sections in this order. This draft targets that format; final page count should be confirmed after student-specific edits.

1. Executive Summary

CloudServe's support team faces a gap between customer demand and timely resolution. The project request was a chatbot, but discovery suggests a safer objective: route each incoming ticket through classification, approved-document retrieval, risk checks and either a sourced draft or an explicit human handoff. The implementation is a local prototype with a FastAPI ticket endpoint, scikit-learn intent and urgency classifiers, TF-IDF retrieval over approved articles, a Chroma vector store, optional Groq/OpenRouter drafting, deterministic checks and SQLite decision logging.

The visible validation set contains 80 tickets and was run once on 2 November 2026. The system produced 8 automatic responses (10%), 65 escalations and 7 blocked outputs. Retrieval found an expected source for 51 of 53 answerable tickets with expected source labels (96.2%). Intent accuracy was 100% on this small labeled set, while urgency accuracy was 43.8%. Intent confidence expected calibration error was 16.6 percentage points, above the 5-point target. Local processing p95 was 6.49 ms, which excludes external model response time and does not represent customer first reply.

These figures describe a conservative but weakly calibrated prototype, not a proven business improvement. Automated checks blocked 7 of 15 generated drafts for grounding concerns. The automated proxy is not human hallucination measurement. No dual-assessor review of at least 50 answers, live first-reply time, live CSAT, or production subgroup response-quality audit has been completed. The hidden 120-ticket assessment set was not available and was not run.

The appropriate decision is to keep the system in a human-supervised evaluation stage. Before any customer-facing pilot, improve urgency performance and calibration, complete independent answer/citation review, instrument live outcome timestamps, validate provider failure handling and review fairness with adequate group sample sizes.

2. Problem

CloudServe Solutions is a business-to-business cloud infrastructure and developer operations company. The supplied brief describes over 500 weekly tickets across email, chat, documentation comments and a community forum, a contractual two-hour response target, observed narrative response times of 8–12 hours, and low first-contact resolution. CloudServe asked for a chatbot. The project brief makes clear that a chat interface alone does not address source quality, uncertainty, accountability, or the queue work that needs human expertise.

The dataset provides a separate synthetic baseline: 219 of 500 development tickets were marked resolved at first contact (43.8%), mean historical CSAT was 2.972/5, and 108/500 had a repeat contact (21.6%). These fields are not live measurements and differ from some narrative figures in the brief. The PRD uses the main project instructions and evaluation framework targets: FCR at least 60%, first reply under five minutes for automated responses, CSAT at least 4/5 and escalation no more than 30%. The contractual two-hour SLA is retained as a separate service obligation; the project brief also contains an inconsistent 65% FCR and 4.2 CSAT variant, which requires assessor clarification.

The product problem is therefore: help agents find and communicate reliable answers faster while protecting customers from unsupported automation, and create a measurable route toward better queue outcomes. A model response is appropriate only when evidence and policy support it; otherwise the system should prepare a concise, useful escalation.

Project boundary

In scope	Out of scope
Ticket normalization, intent/urgency prediction, document retrieval, sourced draft, escalation, guardrails, decision logging and offline evaluation.	Autonomous account changes, autonomous security handling, production deployment, replacement of agents, or claims of live FCR/CSAT/SLA improvement.

3. Discovery Findings

Five stakeholder transcripts were reviewed: Marcus Adeyemi (Head of Support), Sofia Restrepo (Tier One), Daniel Okafor (Tier Two), Ines Carvalho (technical writer) and Ravi Shah (customer). These are fictional capstone evidence, not interviews conducted by the student. They provide useful perspectives but remain qualitative and should not be described as real-world customer research.

Marcus described queue volume above 500 per week, six agents, an 8–12 hour response time against a two-hour commitment, and escalation cost around four times a first-line resolution. Sofia described 40–70 tickets daily, 4–5 minutes for familiar cases and up to 40 minutes for unusual cases; she emphasized poor search and unsafe personal snippets. Daniel estimated that around half of escalations might be resolved by Tier One if agents had the right source and a useful summary. Ines described a 29-article knowledge corpus, weak exact-title search, and risks from unreviewed private snippets. Ravi wanted timely answers that disclose sources and uncertainty.

The development dataset labels 357 of 500 tickets as answerable from documentation (71.4%). That suggests useful corpus coverage, but it does not establish that customers can self-serve, that retrieval yields a safe answer, or that a ticket can be resolved without human review. The dataset's expected route labels mark 311/500 (62.2%) as auto-routable; that label is not actual system success.

The data do not support Sofia's hypothesis that non-fluent customers had worse historical outcomes in development tickets: non-fluent FCR was 45.8% versus 43.2% for fluent customers and historical CSAT was 3.04 versus 2.95. However, these outcomes are synthetic historical fields and say nothing about generated-response quality. In the single validation run, subgroup sizes were small and historical FCR differences were large; subgroup output quality was not human-rated.

Important evidence gaps include first-response timestamps by channel and tier, agent time per task, live customer satisfaction, repeat contact linkage, policy for data retention, and independent quality ratings. Those omissions prevent causal business claims and should be resolved before any pilot.

Discovery measure	Finding	Interpretation
Development tickets	500	Synthetic labeled sample
Channel mix	Email 42.4%; chat 31.0%; docs comments 15.6%; forum 11.0%	Email has the largest share
Historical FCR	43.8%	Dataset baseline; differs from brief narrative ~42%
Historical CSAT	2.972/5	Synthetic rating, not live customer outcome
Documentation answerable	71.4%	Coverage label, not automation success
Mean/median resolution duration	422 / 214 minutes	Resolution time, not first reply

4. Requirements

The PRD v1 was reconstructed as a traceable working document from the supplied pack and current repository. It requires the system to normalize a ticket while preserving its original message and channel; predict intent and urgency separately with alternatives and numeric confidence; retrieve ranked passages whose identifiers resolve to the corpus; answer only from approved evidence; route uncertainty and high risk to a human; validate every generated response; log the decision; and run an unattended evaluation with configurable paths.

Quality requirements include p95 processing under three seconds, safe fallback on provider failure, no secrets in source/history, decision auditability, confidence calibration within five percentage points, and human-rated hallucination no higher than 5% with citation accuracy at least 95% before a pilot. The evaluation framework requires at least 50 answers rated independently by two assessors. This review was not done; those quality thresholds remain unverified.

The implementation target was shaped by the actual evidence: a 71.4% development answerability label supports retrieval as a component, while weak calibration and urgency accuracy argue against broad automatic routing. Requirements that imply customer-visible improvements remain goals to be measured in a live pilot rather than claims about the current prototype.

ID	Requirement	Acceptance evidence
FR-01–02	Normalize tickets and predict intent with ranked alternatives.	API response; per-class precision/recall and confusion matrix.
FR-03	Predict urgency separately; route low-confidence/high-risk cases to humans.	Urgency accuracy and confidence; current accuracy 43.8%.
FR-04	Retrieve relevant approved passages with stable IDs and scores.	51/53 expected source hits in validation.
FR-05–07	Draft only with evidence; block unsafe content; escalate with reason.	8 auto responses, 65 escalations, 7 blocks; human quality review pending.
FR-08–10	Log every decision; support batch harness and kill switch.	80/80 decisions logged; visible harness run.

5. Architecture and Design

The request lifecycle is deliberately small and inspectable. A normalized ticket reaches the classifier; separate intent and urgency models provide scores; retrieval queries the documentation chunks; policy selects automatic response, escalation or block; optional response generation uses the retrieved passages; deterministic validation runs before a draft is returned; and the decision is recorded in SQLite. The FastAPI route is one-ticket-at-a-time. Batch evaluation invokes the same service for each input ticket and saves each decision plus aggregate metrics.

Retrieval uses lexical TF-IDF similarity and a persistent Chroma vector store. It is not a neural embedding model. This choice has low infrastructure cost and is reproducible locally, but lexical retrieval can miss paraphrases and its similarity score is not a calibrated probability of answer relevance. Sources in an API response are the retrieved chunks considered by the policy; they are evidence references, not confidence probabilities.

Ticket classification and urgency use scikit-learn, not an LLM. Optional Groq or OpenRouter generation can draft a response from delimited ticket text and approved source excerpts. The integration uses the Python standard library HTTP client; there is no OpenAI SDK dependency. Provider access was not tested live during this evaluation, so the report does not claim a successful remote generation call. When generation is unavailable, the expected safe behavior is a human handoff.

Confidence scores are classifier outputs and are separate from retrieval similarity. A route threshold is a policy gate. It does not convert either score into proof of truth. Calibration requires comparing predicted probabilities with observed correctness on independent data. The current ECE shows the score should not be used as a calibrated risk probability.

Layer	Component	Role
Ingress	FastAPI /ticket	Validate and accept one normalized ticket per request.
Decision	scikit-learn classifiers	Intent and urgency prediction with alternatives and confidence.
Knowledge	TF-IDF retrieval + Chroma	Rank chunks and return stable source IDs/scores.
Policy	Routing and guardrails	Auto draft only if gates pass; otherwise escalate/block.
Generation	Optional Groq/OpenRouter	Draft grounded response; provider external call unverified.
Persistence	SQLite decision log	Store decisions, sources, reasons, guardrail and timing fields.
Evaluation	Harness	Batch run, metrics and per-ticket outputs.

6. Implementation

The service exposes the /ticket endpoint and returns a ticket ID, action, human-readable reason, intent/urgency prediction and alternatives, retrieved source chunks, optional answer, citations, escalation summary, guardrail results, latency, error and decision ID. An action can be auto_respond, escalate or blocked. A ticket with insufficient intent or urgency confidence is escalated. A filesystem kill-switch file forces an automatic response to escalation.

The intent/urgency models are trained from the development sample. A calibrated classifier was added, but the holdout evidence in this 80-ticket run is not sufficient to establish general calibration, and its measured ECE fails the target. The classifier can return alternatives, which are other candidate labels and their model scores; they are not RAG retrieval results.

The generator is optional. Runtime configuration selects Groq or OpenRouter and reads credentials from environment configuration. The source uses standard-library HTTP rather than a provider SDK. No credential should be included in the submission archive. The source review should confirm the active .env is excluded and that no key is present in Git history before final submission.

The validator performs deterministic PII, instruction-integrity, grounding and tone/scope checks. It can block a response. A lexical grounding check is a safety heuristic, not semantic proof; it can both miss unsupported paraphrases and block acceptable answers. The SQLite record supports audit but introduces data-retention and file-access responsibilities if real customer text is used.

The visible test suite passed 9 of 9 tests during the current implementation session. A clean checkout on a separate machine and a live provider request have not been verified. The exact submission gate still requires the full 80-ticket visible validation set to complete unattended from a clean setup and a hidden 120-ticket assessment by the assessor.

Acceptance behavior	Current status	Evidence / gap
Normalize/route each ticket	Implemented prototype	One ticket per API request; batch harness iterates.
Classify intent and urgency	Implemented; urgency weak	Intent accuracy 100%; urgency accuracy 43.8% on 80 tickets.
Rank source passages	Implemented	Expected source hit rate 96.2% among 53 labeled answerable tickets.
Safe answer or escalation	Implemented with conservative gates	10% auto response; human answer quality not rated.
Log decisions	Implemented	80 of 80 decisions persisted in local run.
Provider outage fallback	Code path present	Live failure path not externally verified.
Clean checkout / hidden run	Not verified	Hidden 120-ticket set unavailable.

7. Evaluation

The visible validation run used 80 tickets from validation_tickets.json on 2 November 2026, once. The local evaluation output is retained in output/validation_run_2026-11-02/metrics.json with per-ticket decisions. Provider keys were blank and the local safe fallback was used. No model or threshold changes were made based on this visible run. The hidden evaluation set of 120 tickets was not provided and was not run. This report must not be represented as hidden-set performance.

The run processed all 80 tickets and logged 80 decisions. Actions were 8 auto responses (10.0%), 65 escalations (81.25%) and 7 blocks (8.75%). Escalation plus block was 90.0%. Against expected routing labels, route accuracy was 47.5%. That result indicates low automatic coverage and imperfect agreement with expected routing; it does not imply actual first-contact resolution.

Intent accuracy was 100% with macro precision/recall of 100% on the labels available in this run. This unusually strong result on a small, likely templated project sample is not evidence of production generalization. The reported confusion matrix is included in Appendix A. Urgency accuracy was 43.75% (35/80); this is inadequate for autonomous prioritization and should keep urgency-dependent handling under human control.

Intent ECE was 0.1664, or 16.64 percentage points, using 80 labeled observations. In the 0.6–0.8 confidence band, 12 predictions averaged 0.738 confidence and observed 100% accuracy; in the 0.8–1.0 band, 68 averaged 0.851 confidence and observed 100% accuracy. The sample overstates accuracy relative to confidence, but the coarse bands and small sample produce noisy estimates. The calibration target of at most five percentage points was not met.

Expected-document hit rate was 51/53 or 96.2% for answerable tickets with expected source labels. This measures whether retrieval returned at least one expected document, not whether the generated prose was supported. The system’s p95 local processing latency was 6.49 ms, mean 4.02 ms and median 3.59 ms. These numbers exclude external provider latency, network, queue time and human response; they do not measure time to first reply.

Among 15 generated drafts, seven were blocked by the lexical grounding validator, a 46.7% automated block proxy. All 15 citation IDs resolved to known sources, while the narrower automated citation-support proxy counted 8/15 (53.3%). These are automated checks and must not be described as hallucination rate or human citation accuracy. The required two-assessor review of at least 50 responses remains outstanding.

Historical validation FCR and CSAT were 46.25% and 2.6/5 respectively; these are supplied history fields, not outcomes caused by this system. Since 72/80 tickets were escalated or blocked, this prototype did not establish an FCR improvement. There are no timestamps to calculate first response and no live customer satisfaction scores. Repeat-contact impact is also unavailable.

Metric	Validation result	Target / reading
Tickets / runs	80 / 1	Visible set used once; hidden 120 not available
Automatic response	8/80 = 10.0%	Coverage is low; safety-first policy
Escalate / block	65 / 7	72/80 human handoffs or blocks
Intent accuracy	80/80 = 100%	Small labeled set; not generalization evidence
Urgency accuracy	35/80 = 43.75%	Insufficient for autonomous priority
Intent ECE	16.64 pp	Target <=5 pp; fails
Expected source hit	51/53 = 96.2%	Retrieval coverage proxy only
Route accuracy	47.5%	Against expected labels
Local processing p95	6.49 ms	Excludes external generation/network
Automated grounding block proxy	7/15 = 46.7%	Not human hallucination rate
Decision log coverage	80/80 = 100%	Local SQLite evaluation log

7.1 Evaluation Method and Metric Definitions

The validation harness processed each ticket through the local service and compared predictions and routing against the supplied labels. Intent accuracy is the share of labeled tickets for which the selected intent matched the provided intent. Precision and recall are calculated per represented class, with macro averages treating each represented class equally. In this run, each represented intent had matching predictions, producing the diagonal confusion matrix. The support per intent ranges from one to six, so one changed prediction would materially alter several class rates.

Urgency accuracy uses the supplied urgency labels and the model urgency prediction. It is reported separately from intent classification because urgency is a distinct task and its failure consequences differ. The observed 35/80 correct predictions mean 45 tickets were assigned a different urgency than the expected label. The current threshold policy did not make this classifier reliable; a human triage path remains necessary.

Expected-document retrieval hit is measured only for tickets labeled answerable with expected document IDs. A ticket counts as a hit if at least one expected document appears in retrieved sources. This does not test whether the retrieved chunk contains enough detail, whether every response claim is supported, or whether the answer would resolve the issue. It should be paired with human relevance judgments and a claim-level citation audit.

Route accuracy compares the system action category with the supplied expected route label. It is not FCR, safe resolution, or a measure of customer outcomes. A conservative escalation can be a safer decision than an expected auto route, while still counting as a route mismatch. The 47.5% result therefore signals disagreement that needs case-level review, not a standalone safety verdict.

Calibration uses expected calibration error (ECE): predictions are grouped into confidence bins and the weighted absolute difference between average stated confidence and empirical correctness is computed. The reported ECE of 0.1664 is 16.64 percentage points. Only two bins have observations in this run; the 80-ticket estimate is sensitive to the sample and bin definition. A larger untouched calibration set, reliability diagram and class-conditional analysis are needed before threshold adjustment.

Processing latency is timed within the local service. It excludes external LLM/provider calls because credentials were blank and local fallback was used, and it excludes queueing, customer delivery and human handling. The p95 value of 6.49 ms should be read as local pipeline compute only. It cannot be compared with the five-minute customer response goal.

Business outcome measures in the dataset are historical labels attached to tickets. They were not caused by this prototype. The validation tickets' historical FCR of 46.25% and CSAT of 2.6/5 are baseline descriptors only. A real pilot would need a pre-registered comparison, a time window, consistent ticket eligibility, response timestamps, customer rating response rate, repeat-contact matching and a method to distinguish system effect from volume or staffing changes.

Table 7.1. Evaluation measurement boundaries

Measure	Numerator / comparison	What it does not establish
Intent accuracy	Correct predicted intent / labeled tickets	Generalization to other populations
Urgency accuracy	Correct urgency / labeled tickets	Severity-weighted harm or triage safety
Expected-source hit	Answerable tickets retrieving >=1 expected doc / eligible tickets	Semantic answer support or correctness
Route accuracy	Action matches expected route / labeled tickets	FCR or customer benefit
ECE	Weighted absolute confidence–accuracy gap by bins	Perfect calibration; subgroup calibration
Local p95	95th percentile local process latency	Customer-visible first reply

7.2 Business Interpretation and Next Evaluation Design

The business target is not merely to answer more tickets. The combined objective is to increase first-contact resolution and customer satisfaction while reducing wait and avoidable escalation without increasing unsupported responses or subgroup harm. The current 10% automatic-response rate means the prototype is not yet positioned to deliver a large queue reduction. If the 8 auto-response tickets were all resolved without human assistance—which this offline harness does not observe—the maximum direct containment in this sample would still be 10%; actual FCR impact is unknown.

The 90% human handoff-or-block share is consistent with a conservative prototype, but it is above the project's <=30% escalation target when blocks and escalations are combined. This is a product trade-off: loosening gates to reach the target before measuring answer safety could exchange visible queue cost for hidden customer harm. First improve model behavior, calibration and source support, then evaluate safe coverage using a holdout set that was not used for model or threshold selection.

A next evaluation should use a development/validation split grouped to avoid near-duplicate leakage, with all thresholds selected only on development data. The untouched assessment set should be processed once at the end, with code version, prompt version, dataset hash, run count, outputs and metrics recorded. Include per-intent precision/recall and confusion matrix; urgency confusion matrix and severity-weighted error; source recall at k; human answer and citation ratings; calibration plot/ECE; route coverage and route error; p50/p95 latency; and subgroup coverage and quality with uncertainty intervals.

For human review, two assessors should independently judge at least 50 generated answers for factual support, citation correctness, completeness, action safety, tone and privacy. Disagreements should be reconciled after independent scoring, and an agreement statistic should be reported. Sampling should include automatically answered and blocked/escalated cases, risk categories and language groups. Since only 15 drafts were generated in this run, it does not meet the required sample size.

Business evaluation requires instrumenting the ticket lifecycle: received timestamp, first substantive reply timestamp, resolution and reopen status, escalation, channel, customer segment, survey invitation and response. A staged pilot should preserve human override, define a stop rule, compare with a contemporaneous control or matched baseline, and report both average outcomes and harmful tails. No target should be claimed as achieved based on the present offline data.

7.3 Acceptance Criteria Review

The build specification treats twelve behaviors as pass/fail criteria. The project implements key paths but the independent clean-checkout gate has not been completed. The status below is an engineering self-review based on source inspection and the visible evaluation run, not an assessor decision. A self-review should not be confused with acceptance by the course evaluator.

Ingestion and normalization are represented by the ticket schema and API path, and the batch run processed all records without dropping a ticket. Classification returns intent and urgency with alternatives and scores, but the measured urgency quality and confidence calibration do not support using the scores as reliable probabilities. Retrieval returns ranked passages with identifiers; expected document recall was high in the sample, but the threshold and semantic relevance need human evaluation.

Routing is deterministic for a fixed model/configuration and records a manager-readable reason. Escalation contains summary and sources considered. Generation uses an optional provider integration with source context, and the guardrail validator can block. However, the provider path was not exercised with a live key in this run, and the response structure, prompt isolation and outage behavior still require clean-machine and adversarial checks. Logs contain decision records, but production privacy controls and retention are not defined.

The unattended run is the most important end-to-end criterion. This environment completed one 80-ticket visible run and produced an aggregate report and per-ticket decision file. The specification also requires an input path and output path interface and the hidden 120-ticket evaluation on a clean checkout. That hidden set was unavailable and no clean external checkout was performed. The project should therefore describe this as a successful local visible run, not as a passed final gate.

Build criterion group	Status	Open proof
Ingest all channels / normalize	Implemented prototype	One-channel-at-a-time external integration not tested
Classify intent + urgency + alternatives	Implemented	Urgency accuracy 43.8%; confidence calibration fails
Retrieve ranked source passages	Implemented	Human relevance not assessed
Deterministic routing + reason	Implemented	Route agreement 47.5%; case review needed
Generate cited answer / safe fallback	Implemented path	Live provider and outage behavior unverified
Validate all replies / block PII and claims	Implemented heuristics	Human semantic audit missing
Unattended run / all tickets logged	Visible local run completed	Clean checkout and hidden 120 run outstanding
Repository credential scan / attribution	Must complete before packaging	Review Git history and archive contents

8. Governance and Risk

The governance design is fail-closed: low confidence, sensitive intents, unsupported source evidence, detected PII or instruction-integrity failures should lead to human review. A filesystem kill switch disables automatic responses. Decisions include action and reason, prediction scores, sources, guardrail results and timing in a local SQLite log. Before real data, the owner must define access controls, retention, deletion, encryption/backups and redaction because the log may contain ticket text.

Fairness analysis reports group counts and historical outcomes separately from system response coverage. In this one validation sample, automated response rates ranged 2.98 percentage points across customer tiers, 10.58 points across regions and 0.69 points by language fluency. These are coverage differences, not answer-quality comparisons. Historical FCR ranges were 24.4 points across tiers, 23.8 across regions and 33.0 across fluency groups. They are based on small samples (enterprise n=8, Latin America n=7, non-fluent n=19) and are not caused by this system. The development sample had no worse non-fluent historical FCR/CSAT pattern, so the combined evidence is mixed.

The project target of less than five percentage points concerns quality variation, but generated response quality was not independently rated. Therefore fairness compliance is unknown. A next-stage review should use two raters, report agreement, sample counts by group, confidence intervals, language coverage and severity-weighted errors. If harmful error or material quality gaps appear, disable automatic responses for that segment while the cause is investigated.

An incident response owner should pause automated responses with the kill switch, preserve decision IDs/logs, identify affected tickets and prompt/model/data versions, notify the support and privacy owners, correct customer impact through a human, assess data exposure, document root cause and approve a monitored re-enable. This is documented in docs/governance.md. Named organizational owners and escalation contacts must be supplied before deployment.

Risk	Severity	Control / current status
Unsupported or harmful answer	High	Grounding gate, conservative threshold, human escalation, independent audit pending.
PII disclosure	Critical	PII guardrail; no detections in validation; production privacy policy absent.
Prompt injection	High	Ticket text treated as untrusted; integrity guardrail and adversarial tests.
Urgency misclassification	High	43.8% accuracy; require human triage; retrain and validate.
Bias/quality gap	High	Coverage and historical metrics only; no human quality ratings by group.
Provider outage	Medium	Optional provider and safe fallback; live failure not tested.
Credential/log exposure	High	Environment secrets; exclude .env; restrict and retain logs deliberately.
Uncontrolled release	High	Versioning, kill switch and rollback procedure; operational ownership needed.

9. Requirements Revision

The Stage 5 revision log records material changes based on the build and visible evaluation. Urgency classification is now explicitly separated from intent and is not allowed to drive autonomous priority when confidence is weak, because accuracy was 43.75%. “Answer when documentation exists” was narrowed to “answer only when classifier, source grounding and safety gates pass”; only 10% auto-responded, and seven of 15 generated drafts were blocked by the grounding validator. A human quality audit is now a prerequisite to any hallucination or citation-quality claim because automated lexical checks cannot establish semantic support.

The confidence requirement was revised from an assumed useful 0.55 threshold to a conservative policy gate that still requires independent calibration evidence; the measured 16.6-point ECE fails the target. The harness requirement emphasizes arbitrary input/output paths for the unseen 120-ticket assessment. Fairness requirements now distinguish historical outcome and routing-coverage differences from generated answer quality. Live outcome claims require timestamped production telemetry.

Two target discrepancies remain open. The main instructions and evaluation framework specify FCR >=60% and CSAT >=4.0, while a Project Brief variation cites FCR >=65% and CSAT >=4.2. This report uses the former as the assessment metric while preserving the inconsistency for the course owner. The contractual two-hour response promise and the under-five-minute automated response target describe different clocks and must be measured separately.

Version 1 assumption	Evidence	Version 2 position
Urgency prediction is adequate for routing.	43.75% validation accuracy.	Keep high-impact urgency decisions with humans.
Confidence threshold means calibrated certainty.	ECE 16.64 pp.	Threshold is a policy gate; calibration remains unmet.
Documentation answerability means safe auto-answer.	Only 8/80 auto; 7/15 drafts grounding-blocked.	Require evidence, confidence and safety checks.
Automated validator proves low hallucination.	No dual-human ratings.	Require independent review of >=50 answers.
Offline latency proves SLA improvement.	No first-reply timestamps.	Instrument real ticket arrival/reply times.

10. Conclusions

The project now contains a working local support-triage prototype, a retrieval-backed response path, deterministic safety checks, an audit record, a kill switch, evaluation tooling and supporting governance artifacts. The single visible 80-ticket run completed with all tickets logged. The prototype shows promising expected-document retrieval coverage but low automated response coverage, poor urgency accuracy, unmet calibration, and a high grounding-block proxy. Those results are reasons to improve and measure the system before use, not to increase automation prematurely.

The highest-priority engineering work is to correct urgency prediction and establish calibration on a separate development split; improve or replace the grounding check and verify semantic support with two independent raters; validate the whole batch harness from a clean checkout; and test provider outage/fallback behavior. The highest-priority operational work is to add first-response and repeat-contact instrumentation, define privacy and retention, recruit independent assessors, and collect adequately sized subgroup quality data.

The assessment pack lists a deadline of 13 September but gives no year. The current preparation date is 2 November 2026. The student should confirm the actual due date with the course owner. The final package also remains incomplete until the student records the required video, supplies actual effort records and personal reflections, and accepts responsibility for the accuracy and authorship of the submitted content.

Readiness decision

Not ready for unattended customer-facing deployment. Suitable only for continued local testing and a supervised evaluation after the outstanding safety and evidence work is completed.

Appendix A. Detailed Evaluation Results

The following details come from output/validation_run_2026-11-02/metrics.json and decisions.json. The 80-ticket run is the visible validation sample and was evaluated once. Per-class scores are only meaningful for classes represented in this sample.

Intent class	Support	Precision	Recall	F1
account_access	4	1.00	1.00	1.00
api_key_issue	3	1.00	1.00	1.00
api_usage_question	4	1.00	1.00	1.00
authentication_failure	5	1.00	1.00	1.00
billing_query	4	1.00	1.00	1.00
compliance_request	4	1.00	1.00	1.00
configuration_help	4	1.00	1.00	1.00
data_export	4	1.00	1.00	1.00
data_residency	6	1.00	1.00	1.00
database_issue	1	1.00	1.00	1.00
deployment_failure	5	1.00	1.00	1.00
feature_request	3	1.00	1.00	1.00
integration_help	3	1.00	1.00	1.00
onboarding	4	1.00	1.00	1.00
performance_degradation	3	1.00	1.00	1.00
quota_or_overage	2	1.00	1.00	1.00
rate_limit	1	1.00	1.00	1.00
rollback_request	6	1.00	1.00	1.00
security_incident	4	1.00	1.00	1.00
sso_configuration	1	1.00	1.00	1.00
unclear_request	6	1.00	1.00	1.00
webhook_issue	3	1.00	1.00	1.00

Table A1. Intent classification metrics on the visible 80-ticket validation run. A perfect score on this small synthetic sample should be interpreted cautiously; supports range from one to six tickets.

Confusion matrix: all 80 labeled tickets fell on the diagonal in this run (predicted intent matched true intent); all off-diagonal cells were zero. Urgency labels did not achieve this performance: 35 of 80 were correct.

Segment	n	Auto-response rate	Historical FCR	Interpretation
Business tier	30	10.0%	53.3%	Historical label; no generated-quality rating
Standard tier	42	9.5%	38.1%	Historical label; no generated-quality rating
Enterprise tier	8	12.5%	62.5%	Small n
Asia Pacific	21	14.3%	52.4%	Historical label
North America	27	3.7%	44.4%	Historical label
Europe	25	12.0%	48.0%	Historical label
Latin America	7	14.3%	28.6%	Small n
Fluent	61	9.8%	54.1%	Historical label

Segment	n	Auto-response rate	Historical FCR	Interpretation
Non-fluent	19	10.5%	21.1%	Small n; historical FCR only

Table A2. Subgroup validation details. Rates are descriptive only; the sample is small and no human assessment of response quality was completed.

Appendix B. Requirements Traceability

The PRD and workbooks provide the complete working traceability tables. This summary maps key requirements to implementation and status.

Requirement	Repository evidence	Evaluation evidence	Status
Ticket endpoint / normalized input	src/main.py, src/service.py, src/schemas.py	One-ticket API behavior; batch run uses 80 inputs	Implemented prototype
Intent/urgency classification	src/classifier.py	Intent 100%; urgency 43.8%	Urgency quality fails readiness
Retrieval and resolvable citations	src/retrieval.py, corpus data	51/53 expected source hit; IDs resolve 15/15	Retrieval evidence favorable; quality unverified
Grounded generation	src/generator.py, prompts/generation_v1.md	Optional provider; live call not tested	Implementation present; live path open
Validation / guardrails	src/validator.py	7 grounding blocks; 0 PII/injection/tone activations	Heuristic checks; human review outstanding
Decision logging	src/storage.py / SQLite	80/80 decisions persisted	Implemented local prototype
Batch metrics	evaluation/harness.py	Single visible 80-ticket run	Hidden 120-ticket run unavailable
Governance/kill switch	docs/governance.md, src/config.py, src/service.py	Manual configuration / code path	Operational test and owner needed

Table B1. Traceability summary. Confirm filenames against final repository before submission.

Appendix C. Governance, Attribution and AI-use Declaration

The risk register and incident procedure are maintained in docs/governance.md and summarized in Section 8. No production customer data was used. The visible datasets and stakeholder transcripts came from the provided fictional capstone pack. The decision database in the evaluation output is local and should not be used for real customer data without privacy controls.

AI-use declaration: Codex was used as an AI assistant to inspect the project instructions and source, help revise implementation and documentation, calculate/interpret evaluation outputs, and draft this report and workbook content. The student is responsible for the accuracy and attribution of submitted content and for adding genuine personal work records and reflection. Code and dependencies should be attributed according to their actual source. No claim is made that the student personally conducted the fictional interviews or performed independent answer scoring.

The archive should exclude .env files, provider keys, local secrets and unnecessary machine-specific runtime data. Before submission, inspect the full source and Git history for credentials; the source archive must preserve repository history while reflecting the current project state.

Known open evidence items

Item	State
Effort log actual dates/hours	Student to provide actual records.
Student reflection/signature	Student must complete.
Human response review	Not performed; two assessors and >=50 answers required.
Hidden 120-ticket evaluation	Unavailable locally; assessor run not performed.
Live provider request	Not verified with a real provider key.
Clean checkout gate	Not verified on a separate machine.
Video	Student records and adds later.

Appendix D. Submission and Video Checklist

The submission guide requires one ZIP named KaushikDutta_Capstone_Submission.zip with exactly four top-level folders: 01_Video, 02_Report, 03_Workbooks and 04_Source_Code. The report PDF must be named KaushikDutta_Capstone_Report.pdf. Each of the five completed workbooks and the effort log must be included; the video is currently a placeholder and must be replaced by the recording before submission.

Video segment	Time	Content
Problem	0–2 min	Client context, user problem, why a chatbot alone is insufficient.
Discovery	2–5 min	Five transcript perspectives, dataset evidence, disagreements and gaps.
System	5–7 min	Architecture, intent/urgency models, retrieval, thresholds, guardrails and log.
Live demo	7–14 min	At least seven minutes: successful answer, escalation, guardrail block, unattended full run.
Results	14–17 min	Business and technical results; limitations and single-run provenance.
Governance	17–18 min	Risks, fairness evidence, human oversight, kill switch and incident response.
Next steps	18–20 min	Improvement plan, pilot gates and open evidence.

Recording requirements: target 20 minutes (allowed 18–22); be visible on camera during the opening and closing; include a live screen demonstration; use clear audio and readable screen; record at 1080p or higher; submit MP4 named KaushikDutta_Capstone_Video.mp4 or include a text file with a shareable link if the file is too large. Demonstrate success, escalation, guardrail block and full unattended run. Do not present the visible 80-ticket run as the hidden 120-ticket assessment.

The pack deadline is “13 September at 23:59” without a year. Confirm the applicable year with the course owner; this package does not infer one.